

Quality Control Standards — Aurora Petrochemical Refinery

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Document ID	APR-QC-001	Revision	2
Plant	Aurora Petrochemical Refinery	Effective Date	20 Mar 2026
Document Owner	K. Iyer, QA Manager	Classification	Controlled — Quality

1. Purpose & Scope

This standard defines product quality control for finished motor gasoline, ultra-low sulfur diesel (ULSD), and jet A produced at Aurora Petrochemical Refinery. Aligns with the Plant Quality Management System and applicable ASTM/EPA fuel specifications.

2. Product Specifications

Product	Key Attributes	Specification Reference
Motor gasoline	RON, MON, RVP, sulfur, benzene	ASTM D2699/D2700, ASTM D5191, EPA Tier 3 (S < 10 ppm).
ULSD	Cetane, sulfur, flash point, lubricity	ASTM D975, sulfur ≤15 ppm, lubricity HFRR ≤520 μm.
Jet A	Flash point, freeze point, JFTOT	ASTM D1655; flash point ≥38 °C; freeze ≤-40 °C.

3. Procedures

3.1 Sampling

Composite sampling per ASTM D4057 at the truck rack and pipeline manifold. Samples retained for 90 days in tamper-evident containers.

3.2 Lab Analysis

Lab performs in-spec analysis under ISO/IEC 17025 accreditation. Out-of-spec results trigger a Hold per QC-APR-OOS-001.

3.3 Certificate of Analysis

Generated automatically from the LIMS for every batch released. Signed electronically by the QA Manager. Cross-references: ASTM methods and refinery batch IDs.

3.4 Trend Review

Weekly SPC review of cetane, sulfur, and RVP. Process Engineering notified when Cpk falls below 1.33.

4. Out-of-Specification Handling

Any OOS result triggers an immediate Hold on the affected lot, a documented investigation, and where applicable, a CAPA. Disposition decisions (Release, Rework, Reject) require QA approval. The investigation is closed in the eQMS with full traceability per the site's records retention policy.

5. Roles & Responsibilities

Quality Assurance (K. Iyer, QA Manager) owns lot release. Operations executes per the recipe and reports deviations. Maintenance supports calibration of QC-critical instruments per the linked maintenance manual (APR-MNT-001). The site safety procedure (APR-SAF-001) applies to all QC laboratory activity.

6. References

- ASTM D4814 — Gasoline Specification
- ASTM D975 — Diesel Specification
- ASTM D1655 — Aviation Turbine Fuel
- EPA Tier 3 Motor Vehicle Emission Standards
- Internal: QC-APR-OOS-001

7. Revision History

Rev	Date	Author	Summary
1	18 Sep 2024	K. Iyer	Initial issue.
2	20 Mar 2026	K. Iyer	Updated specifications and SPC criteria.

Approvals

Role	Name	Signature	Date
Quality Assurance	K. Iyer, QA Manager	_____	20 Mar 2026
Operations / Site Lead	T. Whitfield, Operations Director	_____	20 Mar 2026
Regulatory / Compliance	Site Compliance Officer	_____	20 Mar 2026

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